

A REPORT

ON

Design & Development of low Power, Compact Pulsating Sensor Based Instrument using Open Source EDA Tools

BY

Achyuthan Raghavan

2023B5AA1097P

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Indira Gandhi Centre for Atomic Research-Kalpakkam

A Practice School-I Station of

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(June, 2025)

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Achyuthan Raghavan 2023B5AA1097P Physics and Electronics and Communication Engineering

Prepared in partial fulfillment of the Practice School-I Course Nos. BITS Cxxx

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Key Words: Turbidity, TSL235R, Arduino Mega, Real Time Clock, Open-Source

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Abstract:

This report presents the design and development of a low-power, compact turbidity sensing instrument using open-source EDA tools and microcontroller-based hardware. The system uses a TSL235R light-to-frequency converter to measure scattered light intensity, which correlates with turbidity. An Arduino Mega with built-in hardware timers captures the frequency output, synchronized with a DS3231 Real-Time Clock for precise sampling. A 3D-printed enclosure houses the sensor and an LED light source. Real-time frequency data is displayed on a TFT LCD and a Python-based GUI built with PyQt5 and pyqtgraph. The system is to be calibrated using standard turbidity solutions. A custom PCB is also in the design phase to allow for compactness and multi-channel integration.



Signature(s) of Student(s)

Date 16 July 2025

Signature of PS Faculty

Date

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TABLE OF CONTENTS

1.	INTRODUCTION.....	6
	1.1 Turbidity.....	6
	1.2 Arduino MEGA and its TIMERS.....	6
	1.3 Real Time Clock and DS3231 Module.....	7
	1.4 TSL235R Light-to-Frequency Converter.....	8
2.	OBJECTIVE.....	10
3.	THE TURBIDITY SENSOR.....	11
4.	METHODOLOGY.....	19
	4.1 Flowchart.....	19
	4.2 3D Modelling and Printing.....	20
	4.3 Frequency Measurement using Arduino Mega.....	21
	4.4 Integration with TFT Display and GUI.....	22
	4.5 TSL235R and LED Integration.....	23
	4.6 Schematic and PCB Design.....	24
	4.7 Testing and Calibration.....	28
5.	ANALYSIS.....	30
	5.1 Frequency Measurement using Arduino Mega and RTC.....	30
	5.2 Working Physics Behind the Turbidity Sensor.....	32
6.	CONCLUSION.....	33
7.	REFERENCES.....	34

INTRODUCTION

1.1 *Turbidity :*

Definition and Causes of Turbidity

Turbidity means the cloudiness or haziness of a fluid due to suspended solids that are usually invisible to the naked eye. These particles scatter and absorb light instead of letting it pass straight through, which reduces water clarity. Common causes of turbidity include natural processes like erosion, the breakdown of organic matter, and biological activity such as algae growth. Human activities such as construction, mining, agriculture, and wastewater discharge can greatly increase turbidity in natural water bodies.

Measurement and Units

Turbidity is often measured using optical instruments called turbidimeters or nephelometers, which evaluate how light scatters in a water sample. The most common unit for measuring turbidity is the Nephelometric Turbidity Unit (NTU). In some cases, other units like Formazin Turbidity Units (FTU) or Jackson Turbidity Units (JTU) may be used. High turbidity readings usually show a higher concentration of suspended solids, although the actual makeup—like clay, silt, organic matter, or microorganisms. Regular turbidity testing is crucial in water treatment facilities to make sure water meets safety and appearance standards.

Environmental and Health Significance

Turbidity is an important sign of water quality. In natural ecosystems, too much turbidity can block light, which affects how aquatic plants photosynthesize and disrupts food chains in the water. Suspended particles can also carry pollutants, heavy metals, or germs, which can create serious health risks if consumed. In drinking water systems, high turbidity can protect harmful germs from disinfection methods, increasing the chance of waterborne diseases. Because of this, regulatory bodies like the World Health Organization set strict turbidity limits for safe drinking water. Knowing how to manage turbidity is essential for protecting the environment, public health, and treating water effectively.

1.2 *Arduino Mega and its TIMERS:*

Introduction to Arduino Mega

The **Arduino Mega 2560** is a powerful microcontroller board based on the **ATmega2560** microchip. It is widely used in electronics projects and prototyping due to its large number of I/O pins,

robust memory, and enhanced capabilities compared to standard Arduino boards like the Uno. It features **54 digital I/O pins**, **16 analog inputs**, **4 hardware serial ports (UARTs)**, **256 KB of flash memory**, and **16 MHz clock speed**, making it ideal for complex projects that require more resources, such as robotics, data logging, or real-time control systems.

Timers and Internal Clock in Arduino Mega

The Arduino Mega comes equipped with **six hardware timers: Timer0, Timer1, Timer2, Timer3, Timer4, and Timer5**. These timers are integral to time-based operations such as **PWM (Pulse Width Modulation)**, **delays**, **measuring time intervals**, and **interrupt handling**. Each timer has a specific bit-width (8-bit or 16-bit) and can operate independently, allowing multiple timing operations to run simultaneously.

Underlying all timing functions is the **internal clock**, driven by a **16 MHz crystal oscillator**. This clock provides the base frequency for the timers and determines how fast the microcontroller executes instructions. Timers divide this base frequency using **prescalers**, allowing them to count at slower, more useful rates for various applications. The precision and flexibility of the internal timers make them essential for tasks that require accurate timekeeping or signal generation.

1.3 Real Time Clock and DS3231 Module:

. Introduction to Real-Time Clocks (RTC)

A Real-Time Clock or RTC is an application-specific integrated circuit that is used to keep track of time and date in embedded devices. Unlike the software implementation of timekeeping, which depends on the system microcontroller and is vulnerable to reset or power loss, RTCs are independent and are usually powered by a battery. This enables them to hold time information even when the main system is turned off. RTCs are typically used in devices like digital clocks, data loggers, time-stamping systems, and scheduling devices.

Regular RTCs monitor time parameters like seconds, minutes, hours, day of the week, date, month, and year, usually with internal leap year compensation. The majority of RTCs interface with microcontrollers using serial communication protocols, the most common of which is I²C, because it is simple and requires few pins.

DS3231 RTC Module

The DS3231 is a highly accurate temperature-compensated Real-Time Clock from Maxim Integrated. It is a big step forward from previous RTCs such as the DS1307, mainly because it has a built-in

temperature-compensated crystal oscillator (TCXO) that provides consistent accuracy over a broad temperature range. The DS3231 keeps time with a precision of ± 2 parts per million (ppm), equivalent to a deviation of less than a minute per year under normal conditions.

Key Features:

- On-chip 32.768 kHz crystal oscillator with temperature compensation
- Timekeeping accuracy of ± 2 ppm from 0°C to +40°C
- Functionality over a range of -40°C to +85°C temperatures
- Keeps time while powered off by using a coin-cell battery
- Two programmable time-of-day alarms
- A square-wave output at user-selectable frequencies (1Hz to 32kHz)
- I²C interface (7-bit address: 0x68) for microcontroller communication

DS3231 normally comes in the breakout module form, which might also contain other components like a backup battery holder and EEPROM. Because of its precision, simplicity of incorporation, and dependability, the DS3231 finds broad application in precision timing among consumer, industrial, and scientific markets.

1.4 TSL235R Light-to-Frequency Converter:

The TSL235R is a very compact and versatile light-to-frequency converter developed and produced by Texas Advanced Optoelectronic Solutions (TAOS). The TSL235R is commonly employed in optical sensing applications where there is a need to detect and measure the amount of light intensity. The TSL235R generates incident light into a frequency output that directly relates to the light intensity, which is particularly suitable for connection to digital systems like microcontrollers, counters, and timers.

Operating Principle

The TSL235R combines a photodiode and a current-to-frequency converter into one monolithic CMOS IC. Upon exposure to light, the photodiode produces a photocurrent that is proportional to the intensity of the light. This current is internally conditioned and developed into a square-wave output signal whose frequency varies linearly with light input. The output signal can be readily read from any digital input that can count pulses or measure frequency.

Key Features

- Broad Spectral Response: 320 nm to 1050 nm, including visible and near-infrared light.
- Linear Frequency Output: The Frequency of output is in direct proportion to the intensity of light.
- Easy Digital Interface: No ADC is needed; output can be read by pulse counting.
- Operating Voltage: 2.7 V to 5.5 V, compatible with 3.3 V and 5 V systems.
- Output Range: Up to 100 kHz, depending on light incidence.
- High Precision and Stability: Aged for reliable and reproducible operation.
- Temperature Range: -25°C to $+70^{\circ}\text{C}$, for use in indoor and moderate outdoor applications.

Applications

The TSL235R is applied in a range of applications, such as:

- Ambient light detection
- Laboratory or agricultural light intensity monitoring
- Optical pulse counting
- Colorimetry and spectrophotometry (when combined with filters)
- Solar radiation measurement
- Light-based RPM measurement or feedback control systems

Its small 3-lead TO-92 package, low power dissipation, and direct digital output make it a great candidate for embedded and battery-operated systems.

Integration with Microcontrollers

Since it has a digital frequency output, it is simple to interface the TSL235R with the input capture or interrupt pin of microcontrollers like those of the Arduino, ESP32, or STM32 families. Through the measurement of the time between rising edges or pulse counting within a known time window, the incident level of light can be accurately ascertained without analog signal conditioning or ADCs.

OBJECTIVE

This paper is a report on the development of a custom turbidity sensor using a 3D-printed housing, a TSL235R light-to-frequency converter, an LED light source, and an Arduino-based microcontroller system. The sensor system aims to measure the intensity of scattered light as a function of turbidity, with frequency output captured using internal Arduino timers.

THE TURBIDITY SENSOR

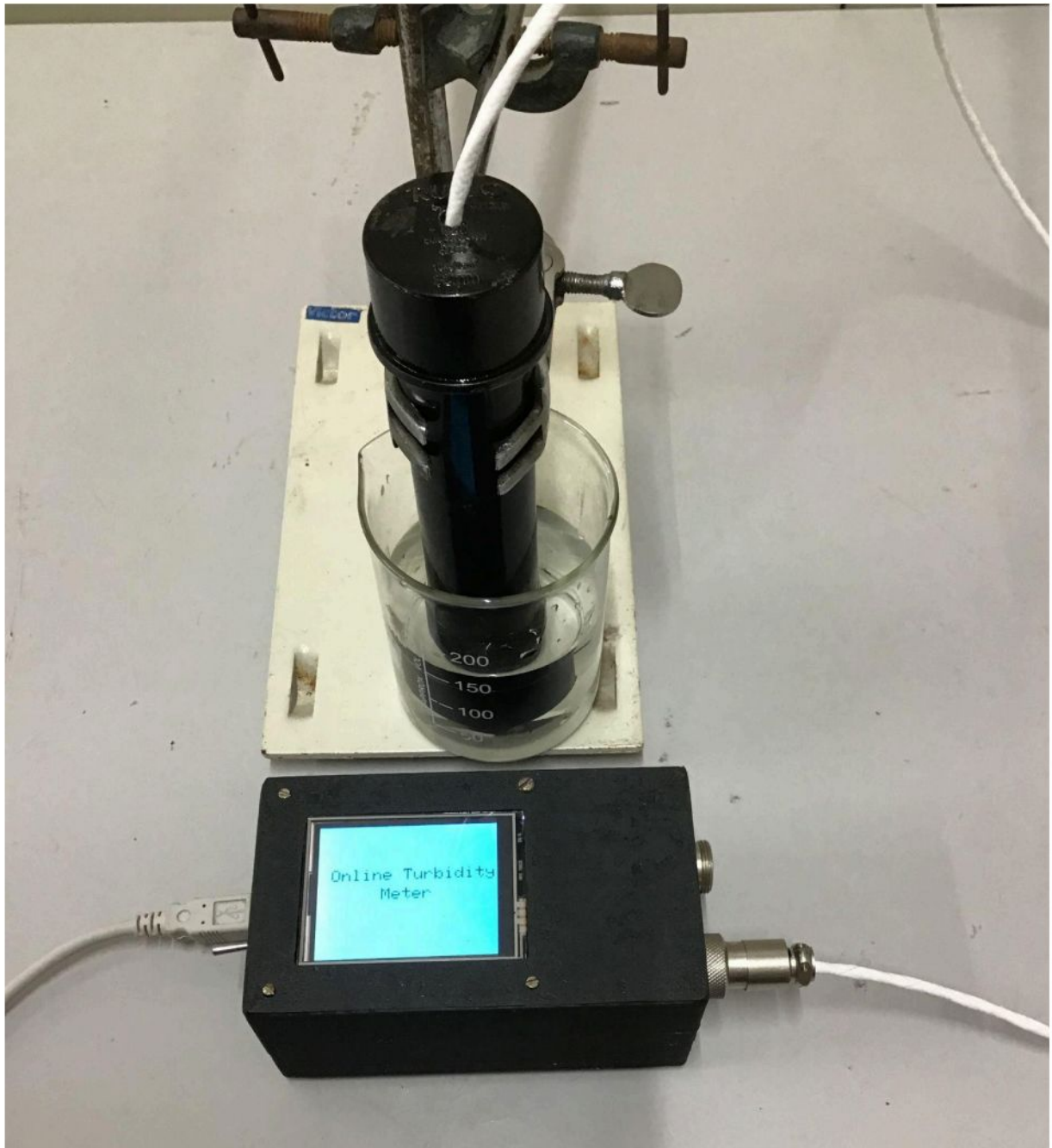


Fig. 1: Turbidity Sensor

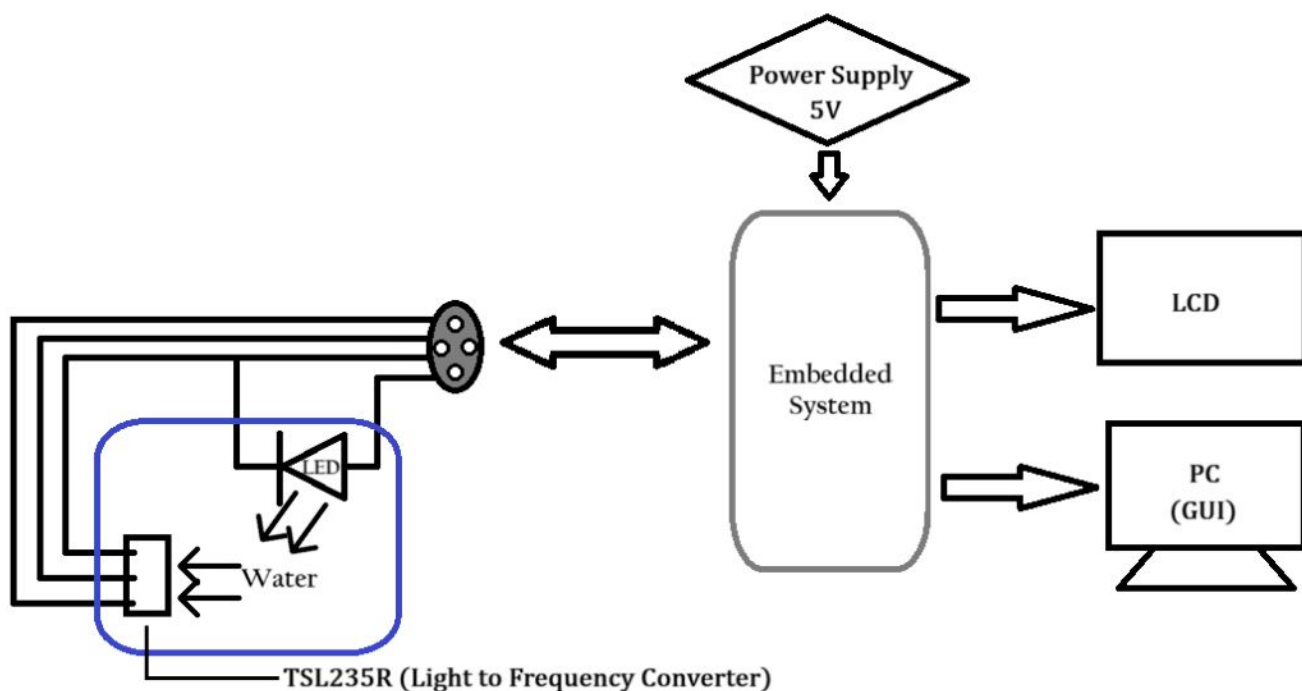


Fig. 2: Block Diagram of the Turbidity Sensor

The Turbidity Sensor and Embedded Systems box, developed at IGCAR is shown in Fig. 1. The Turbidity Sensor consists of the Main sensor, the embedded system, the displays, and the Power Supply, as shown in Fig. 2.

The Main Sensor :

The sensor is made up using an LED and TSL235R placed at a 90-degree angle to each other. The light from the LED is scattered by the particles in the water, and then the Light to Frequency Converter gives out a square wave of frequency linearly proportional to the intensity of scattered light.

The LED:

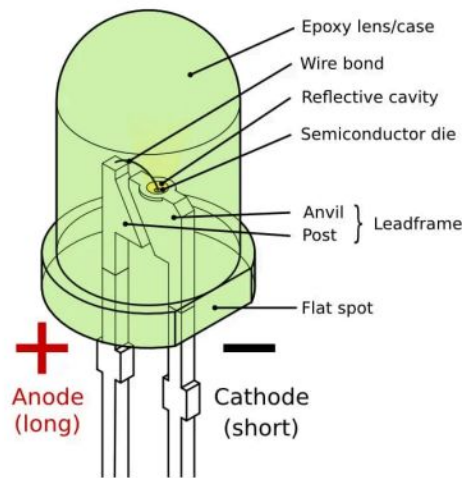


Fig. 3: LED Diagram

An LED (Light Emitting Diode) is a type of diode that emits light when current flows through it. An LED has 2 leads and is made up of an epoxy case, refer Fig. 3. Like other diodes, it only allows current to pass in one direction—from the positive side (anode) to the negative side (cathode). LEDs are made from special semiconductor materials that can emit light when energized.

Inside the LED, there is a junction formed between two types of semiconductor materials: p-type (with holes) and n-type (with electrons). When current flows through the LED in the forward direction, electrons and holes move toward the junction and recombine. This recombination releases energy in the form of light, which is called electroluminescence.

The color of the light depends on the material used and the energy released during recombination. For this project, a white LED is used, which typically uses gallium nitride (GaN) coated with a phosphor layer to produce white light. LEDs are highly efficient, converting most of the electrical energy into light with minimal heat, making them ideal for modern electronic applications.

The TSL235R (Light-to-Frequency Converter) :

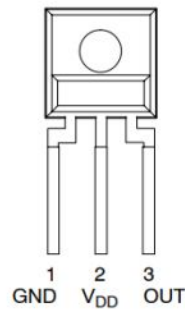


Fig. 4: TSL235R Light-to-Frequency Converter

The TSL235R (shown in Fig. 4), uses a photodiode to convert incident light into current. This current is converted to Voltage using a transimpedance amplifier, which is an op-amp circuit, then the Voltage is converted to Frequency by a voltage-to-frequency converter circuit.

The photodiode is explained in section 4.2.1 of this report

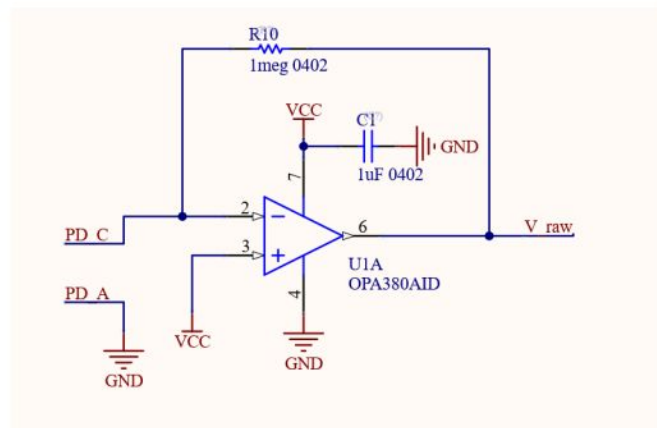


Fig. 5(A): Photodiode Current to Voltage Converter

The Circuit diagram of the Photodiode Current to Voltage Converter is shown in Fig. 5(A).

The photodiode generates a current (I_{ph}) when light hits it, but this current is extremely small (nA - mA range), so we can't use this directly

To measure this current, the circuit uses an **op-amp** with a resistor in its feedback loop. This setup is called a **transimpedance amplifier**.

It takes the **current** from the photodiode and turns it into a **voltage** using **Ohm's Law**:

$$V_{out} = -I_{ph} \times R_f$$

The minus sign is because the current is flowing into the inverting input.

The voltage signal is then sent to a **voltage-to-frequency converter**, made of op-amps and a transistor. This part turns the voltage into a **square wave signal**, where the frequency (how fast the wave repeats) depends on the voltage.

Voltage to Frequency Converter (Square Wave Generator):

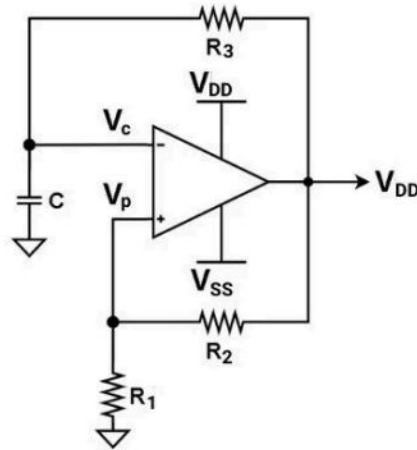


Fig.5(B): Square wave Generator Circuit

The circuit for the square wave Generator is shown in Fig. 5(B).

1. Integrator (Op-Amp 1):

This turns the input voltage into a **ramp signal** (like a line that goes up or down steadily).

Higher voltage → steeper ramp (goes up/down faster).

Formula:

$$V_{out}(t) = - (1 / RC) \times \int V_{in} dt$$

2. Comparator Stage (Op-Amp 2)

The second Op-Amp watches the ramp signal.

When the ramp hits a certain voltage ($\pm V_{ref}$), the comparator **switches the output** (from HIGH to LOW or LOW to HIGH).

3. Transistor Reset Switch

Every time the comparator switches, it tells the transistor to **reset the ramp** by discharging the capacitor.

This causes the ramp to start again in the opposite direction.

How Frequency Is Determined

The slope of the ramp depends on the input voltage.

The faster the ramp reaches the reference voltage, the sooner the comparator flips and triggers a new cycle.

Therefore, higher input voltage → faster ramp → higher frequency.

This gives the final relationship:

$$f_{\text{out}} \propto V_{\text{in}}$$

Functional Block Diagram

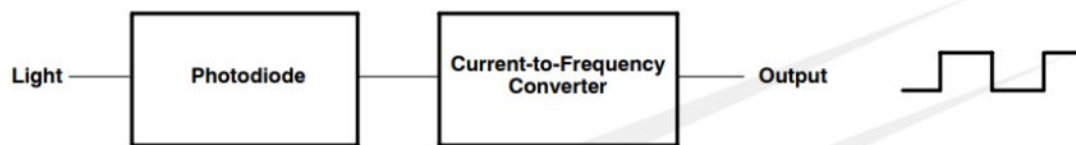


Fig. 5(C): Block Diagram of TSL235R

Fig. 5(C) shows the overall working of the TSL235R.

The Embedded System:

The embedded system consists of an Arduino Mega (microcontroller), Real time clock (DS3231), a Timer, and an Interrupt System (software). All of these components are discussed in the Introduction as well as sections 3.3 and 4.1. Along with these, the power supply and LCD display are also part of the embedded system.

Power Supply:

The 5V power supply is connected directly to the Arduino Mega, which in turn powers the microcontroller along with the LED, TSL235R, and LCD.

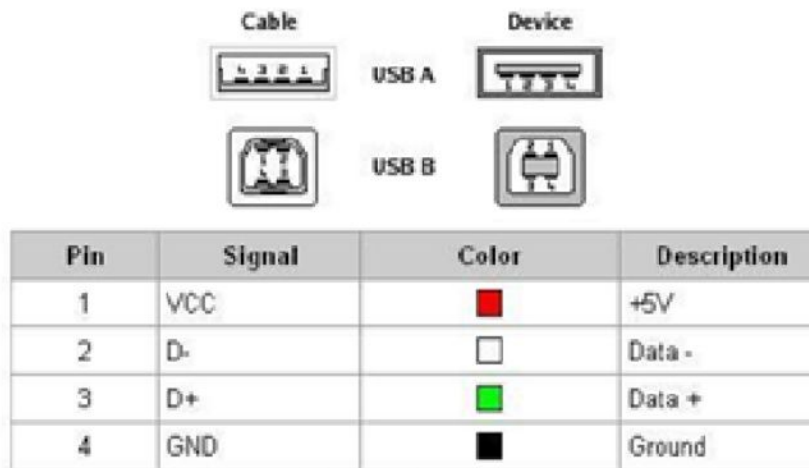


Fig. 6: USB A and USB Cable

The power Supply cable is a USB A to USB B Cable as shown in Fig. 6.

The LCD Display:

The turbidity and frequency values will be displayed on a TFT LCD (Fig. 7(A)) along with the time and date given by the RTC.



Fig. 7(A) TFT LCD

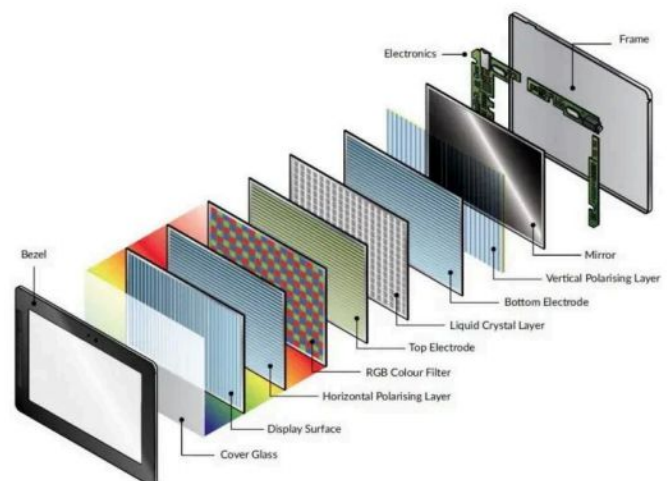


Fig. 7(B) Layers of the TFT LCD

As seen in Fig. 7(B), a TFT LCD is made up of multiple layers. TFT LCD stands for **Thin-Film-Transistor Liquid-Crystal Display**. Each pixel is controlled by a separate thin-film transistor, which acts like a tiny

electronic switch. These transistors are arranged in a grid on a thin glass layer, and they control how much light passes through each pixel.

The screen works by using **liquid crystals** that can change how they align when an electric field is applied. These crystals are placed between two polarizing filters and aligned in such a way that, when no voltage is applied, light can pass through them. When a voltage is applied to a specific pixel using its transistor, the liquid crystals twist and block or allow light to pass in different ways. This creates the image you see on the screen.

Behind the liquid crystal layer is a **backlight**, usually made of white LEDs, which provides the light source for the display. The liquid crystals themselves don't emit light — they only control how much of the backlight passes through. Color is added using **RGB color filters** (Red, Green, Blue), so each pixel actually contains three sub-pixels. By adjusting the voltage to each sub-pixel, different colors and brightness levels can be displayed.

METHODOLOGY

4.1 Flowchart:

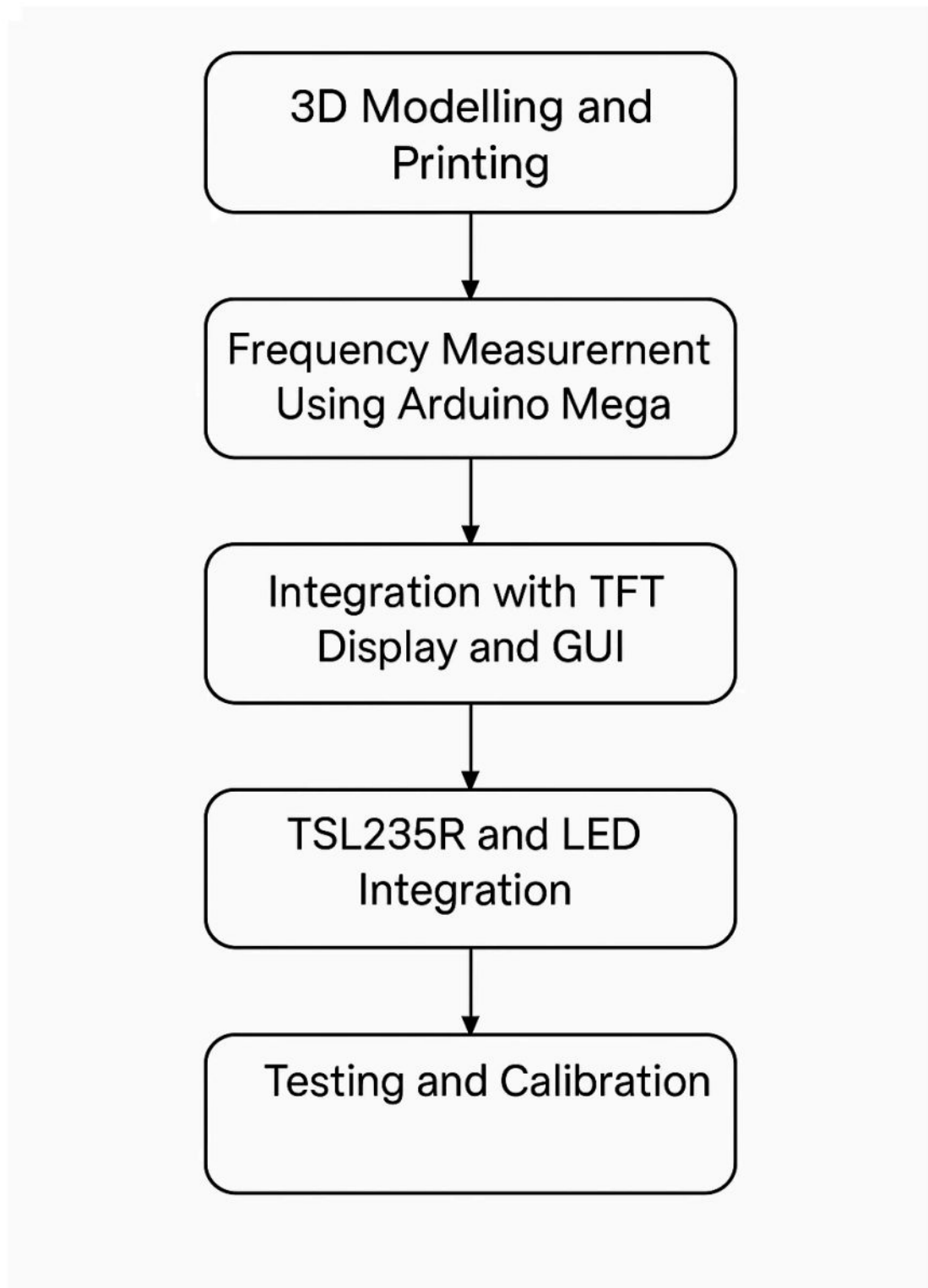


Fig. 8: Flowchart of Methodology

The overview of the methodology is shown as a Flowchart in Fig. 8.

4.2 3D Modelling and Printing:

The outer casing of the turbidity sensor was designed using TinkerCad and Fusion 360, shown in Fig. 9(A) and Fig. 9(B). Once finalized, the model was 3D printed using PLA filament, ensuring proper alignment for sensor components and durability for handling.

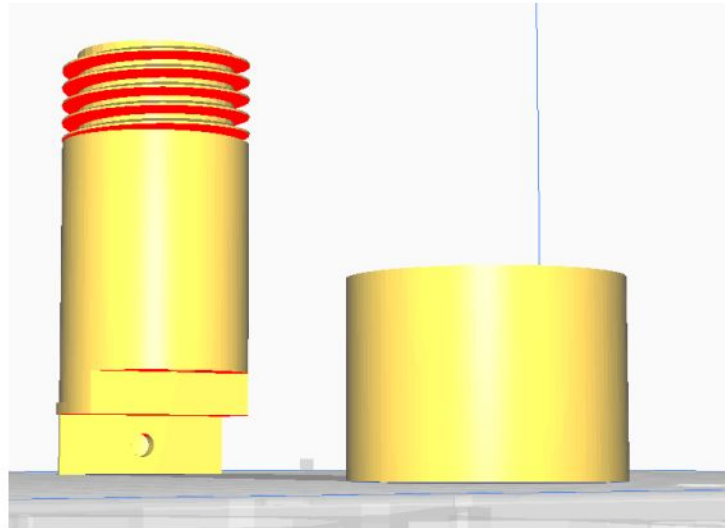


Fig. 9(A): 3D Model (A)

The Fig. 8 shows the placement of the LED and TSL235R in the 3D print

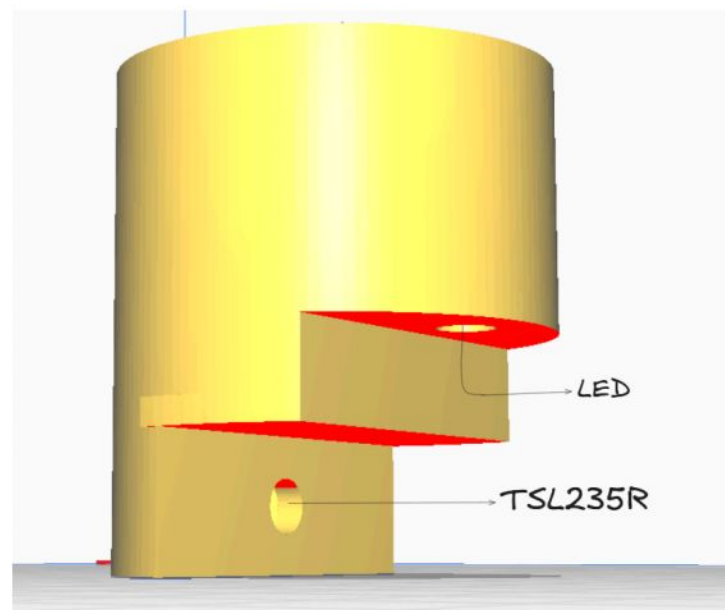


Fig. 9(B): 3D Model (B)

4.3 Frequency Measurement Using Arduino Mega:

The setup uses Arduino Mega's hardware timers to count how many pulses occur on each channel between 1-second square wave signals from the RTC. Overflow counters extend the timer range. When an interrupt triggers, the counts are captured, the timers are reset, and the frequency is calculated. This Freq calculated is then displayed as shown in Fig. 10.



Fig. 10: Square Wave input from Wave Generator Displayed as Frequency on LCD

4.4 Integration with TFT Display and GUI:

The real-time Turbidity values were displayed on a TFT LCD screen using the MCUFRIEND_kbv library (as shown in Fig. 12). Additionally, a Python-based GUI (as shown in Fig. 11) was developed using PyQt5 and pyqtgraph on Visual Studio Code, which plots the Turbidity in real time via serial communication. The GUI has option to export the data received to a CSV file for logging.

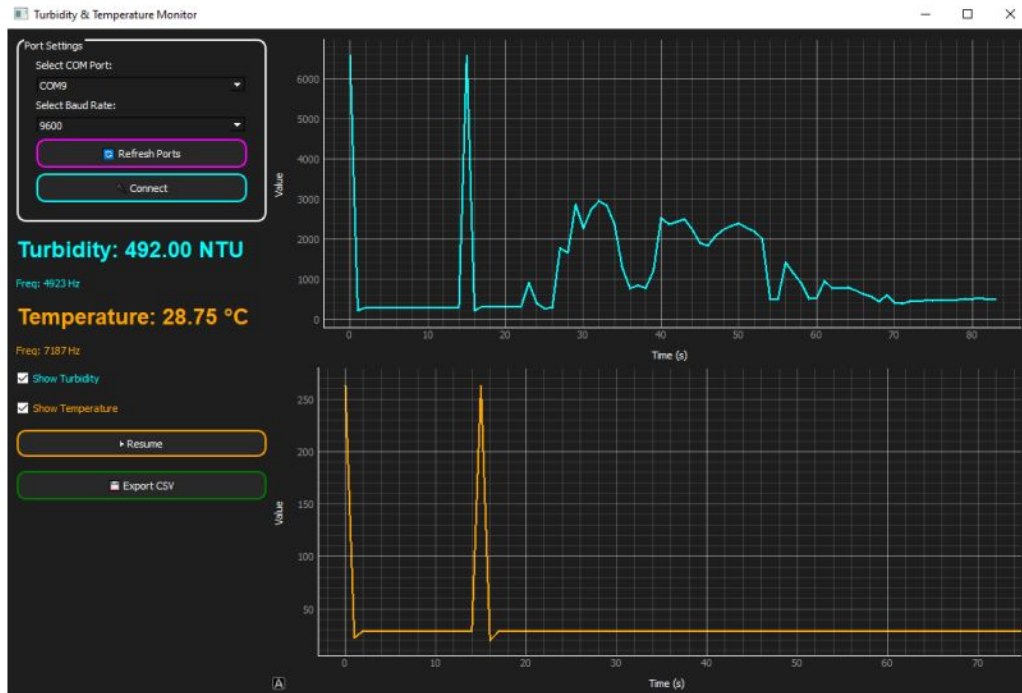


Fig. 11: Graphical User Interface (GUI)

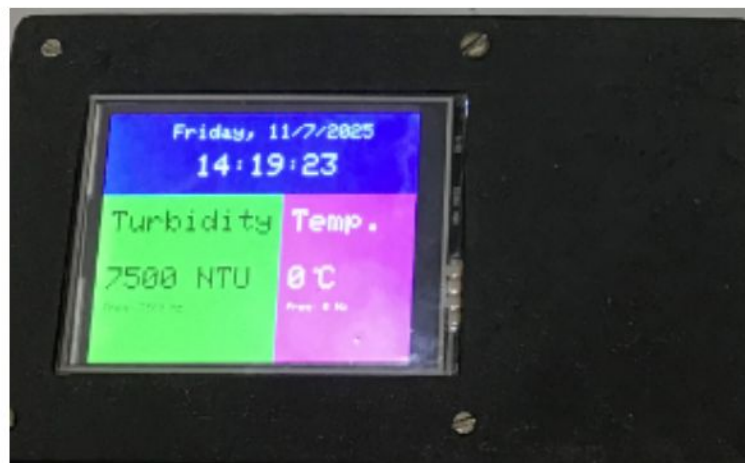


Fig. 12: TFT LCD Display

4.5 TSL235R and LED Integration:

To complete the turbidity sensor, the TSL235R light-to-frequency converter and an LED were mounted into their respective slots in the 3D printed enclosure as shown in Fig 13. These were then electrically connected to the Arduino Mega for data acquisition as shown in Fig. 14 and Fig. 15.



Fig. 13: Prototype Turbidity Sensor

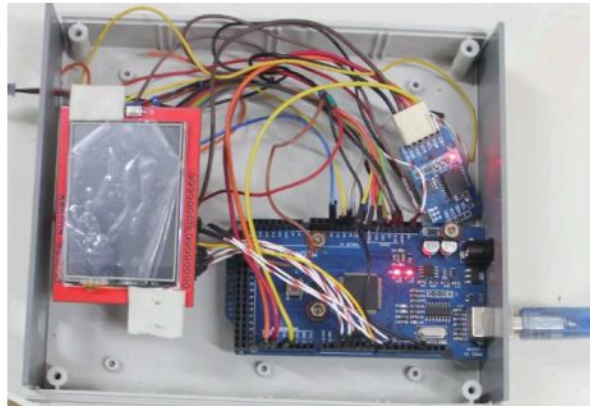


Fig. 14: Internal Circuitry of the prototype Sensor

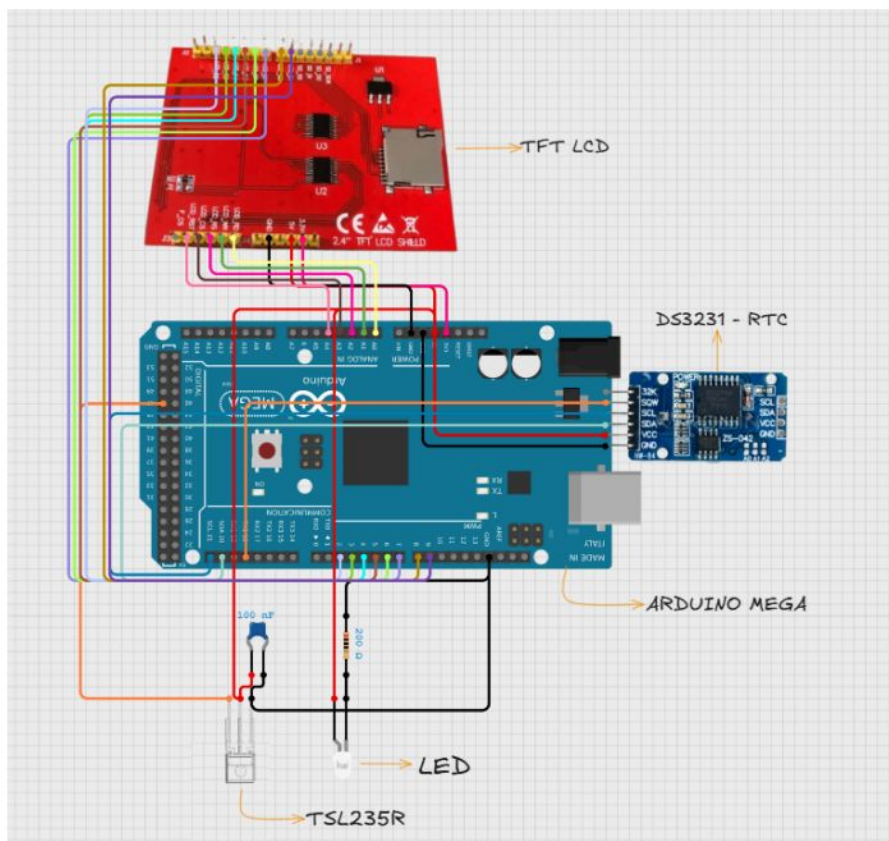


Fig. 15: Schematic of the Internal Circuitry

4.6 Schematic and PCB Design:

To make the sensor more compact and robust, a custom Schematic and PCB (refer to Fig. 14 and Fig.)have been designed on KiCAD 9.0, an Open-Source Software. The PCB is designed using the ATmega 2560 microcontroller chip used in the Arduino Mega.

Additionally, to add up to 4 channels, the 4 free TIMER Pins of the ATMEGA 2560 must be accessed; only 1 such pin is accessible in the Arduino Mega Development Board.

The Schematic Designed is Shown in Figures : Fig. 16(A)-(D), the PCB is shown in Fig. 17 and the 3D models of the PCB is shown in Fig. 18(A) and Fig. 18(B)

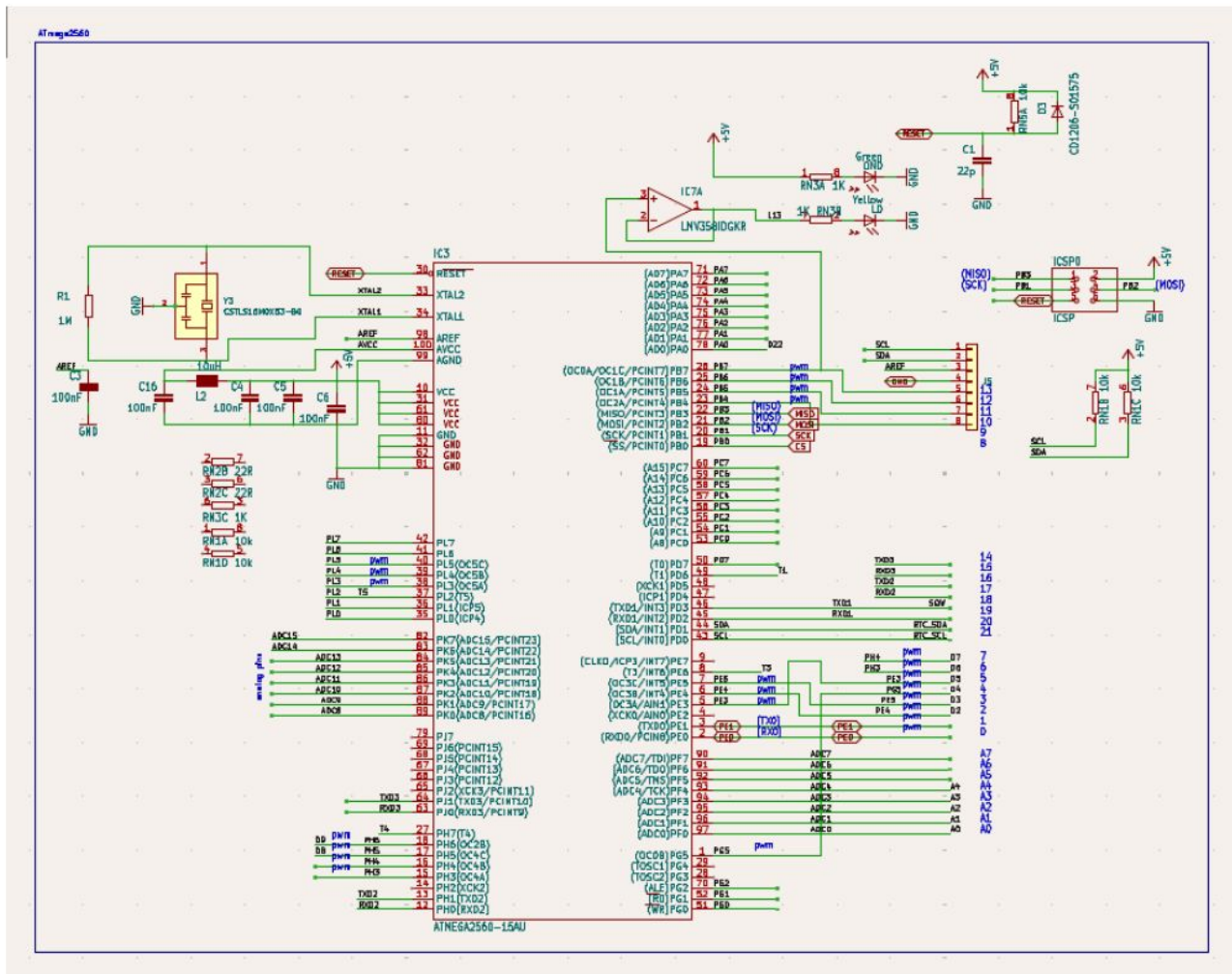


Fig. 16(A): Schematic Design of the Custom PCB(A)

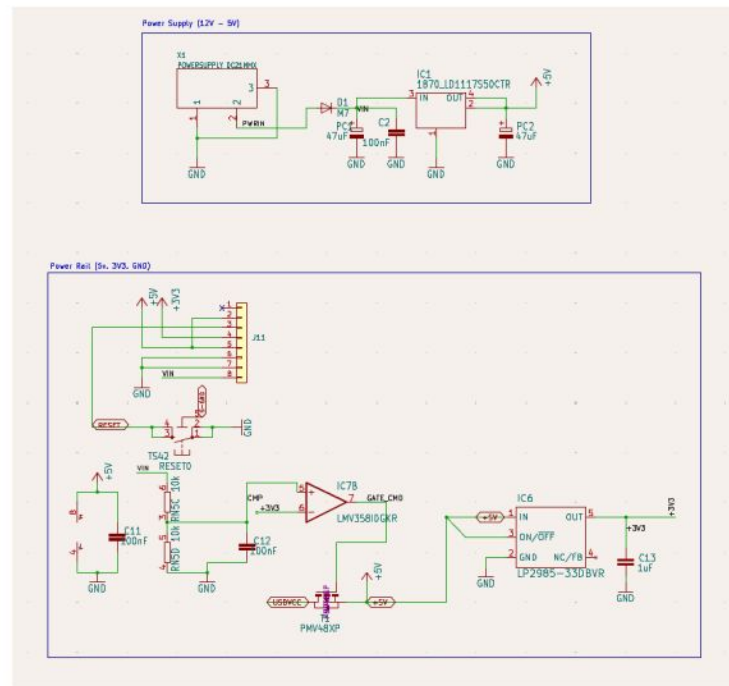


Fig. 16(B): Schematic Design of the Custom PCB(B)

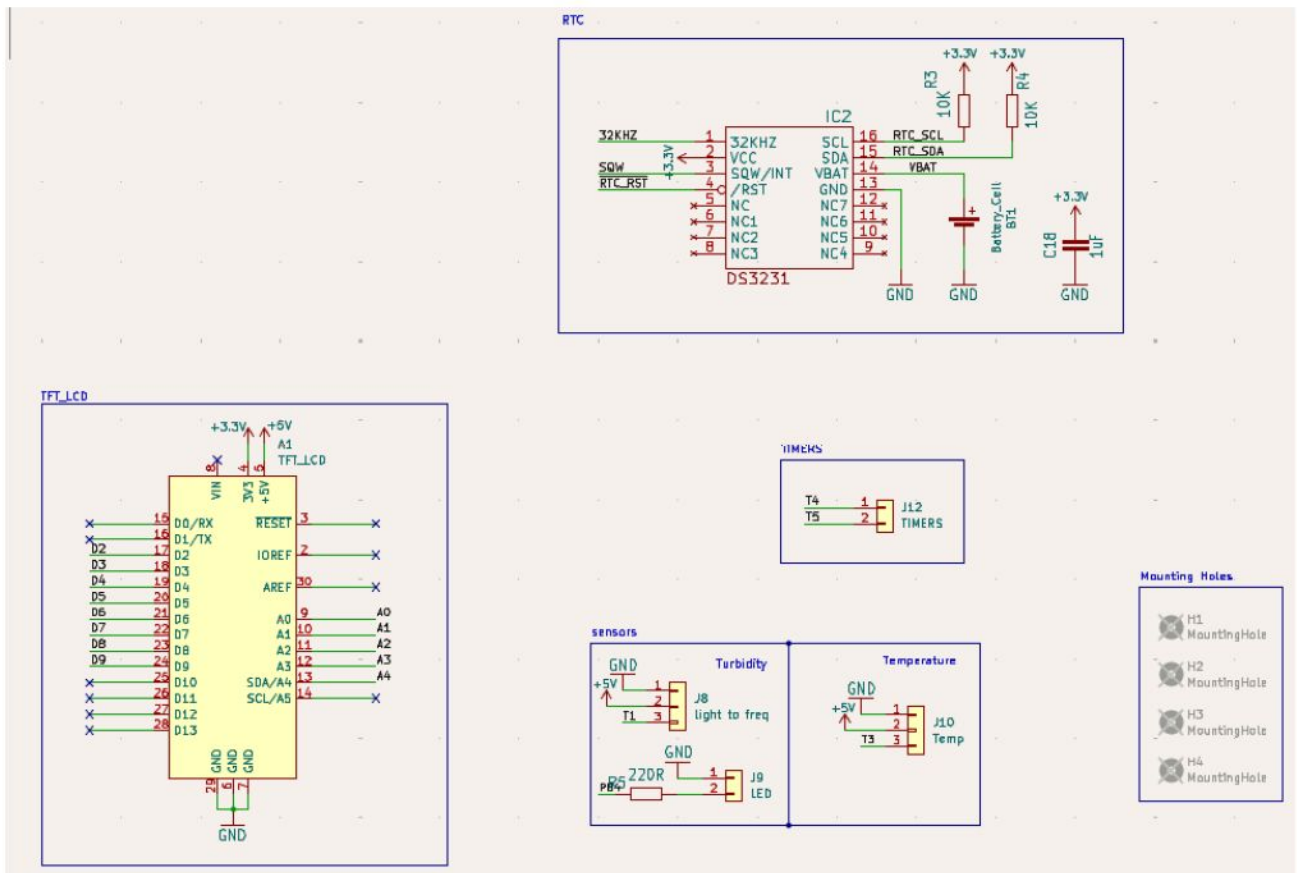


Fig. 16(C): Schematic Design of the Custom PCB(C)

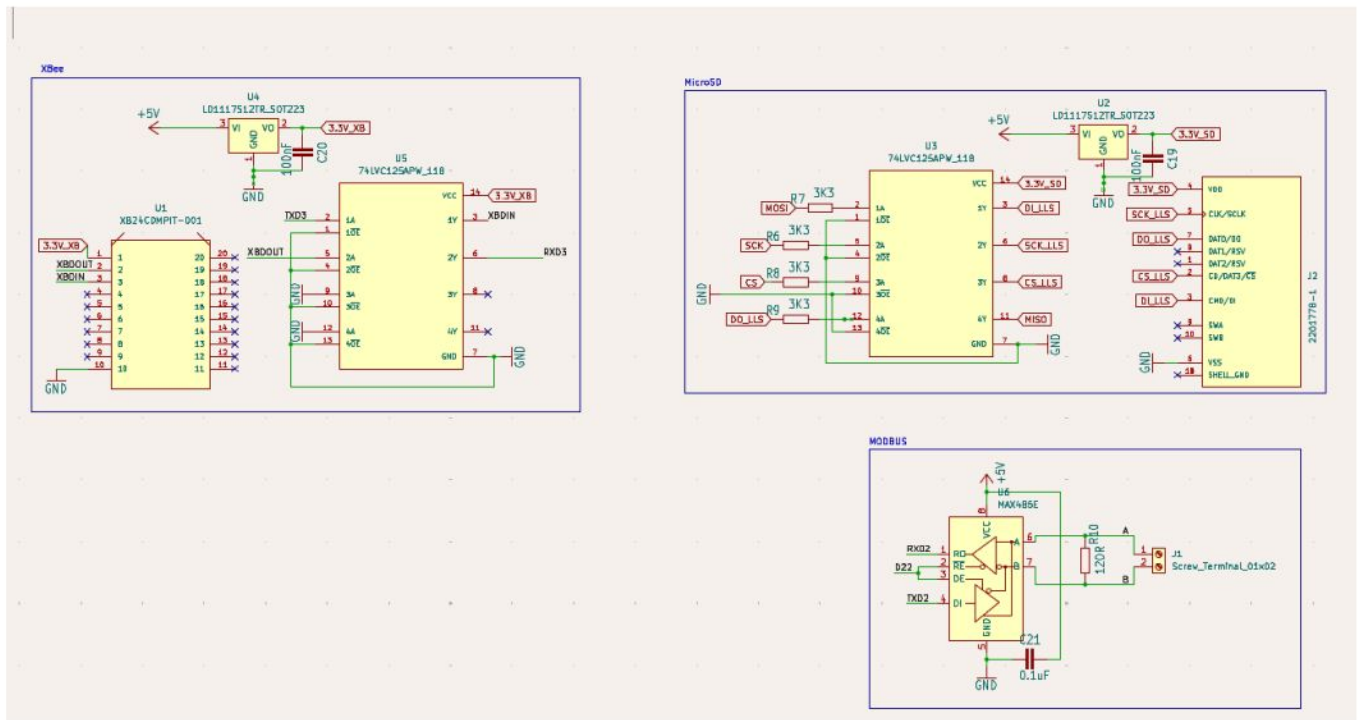


Fig. 16(D): Schematic Design of the Custom PCB(D)

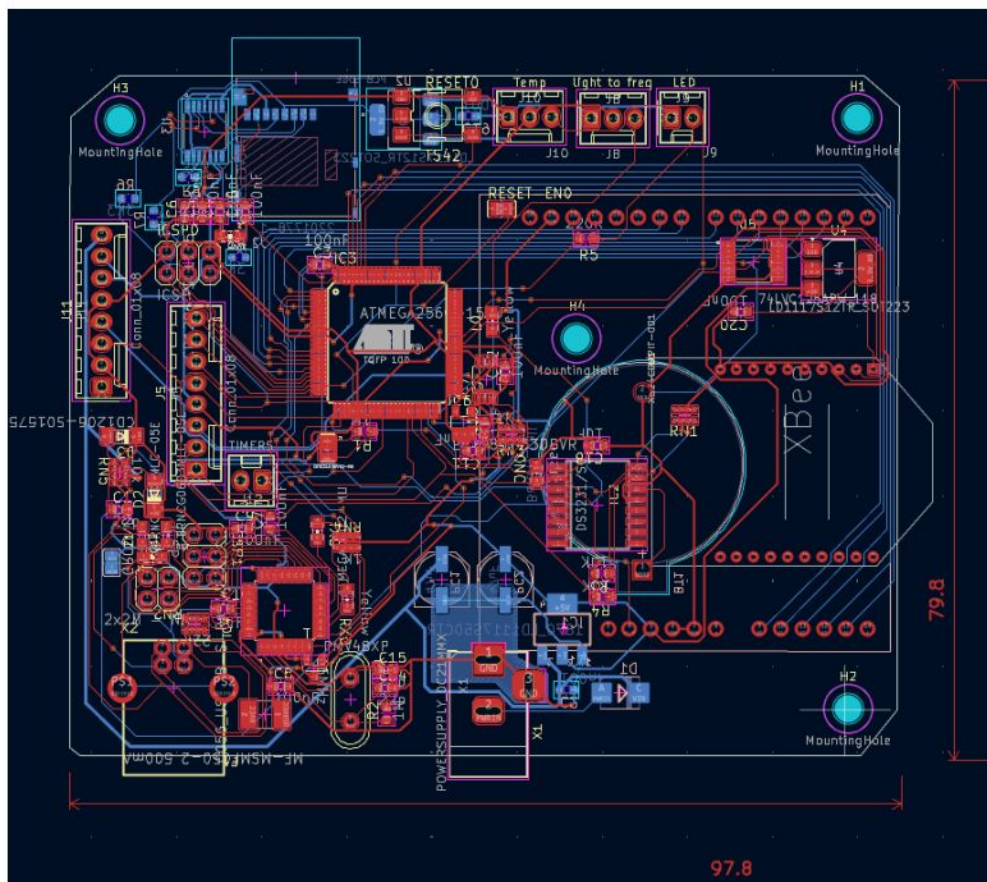


Fig. 17: Custom PCB Design

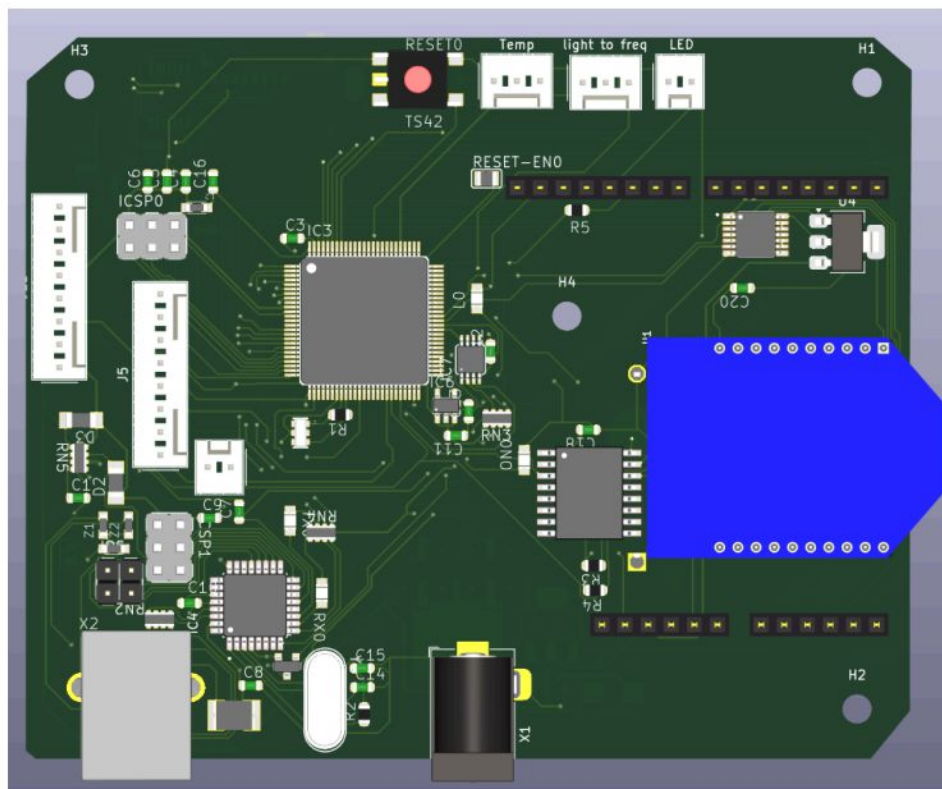


Fig. 18(A): 3D Model of PCB(A)

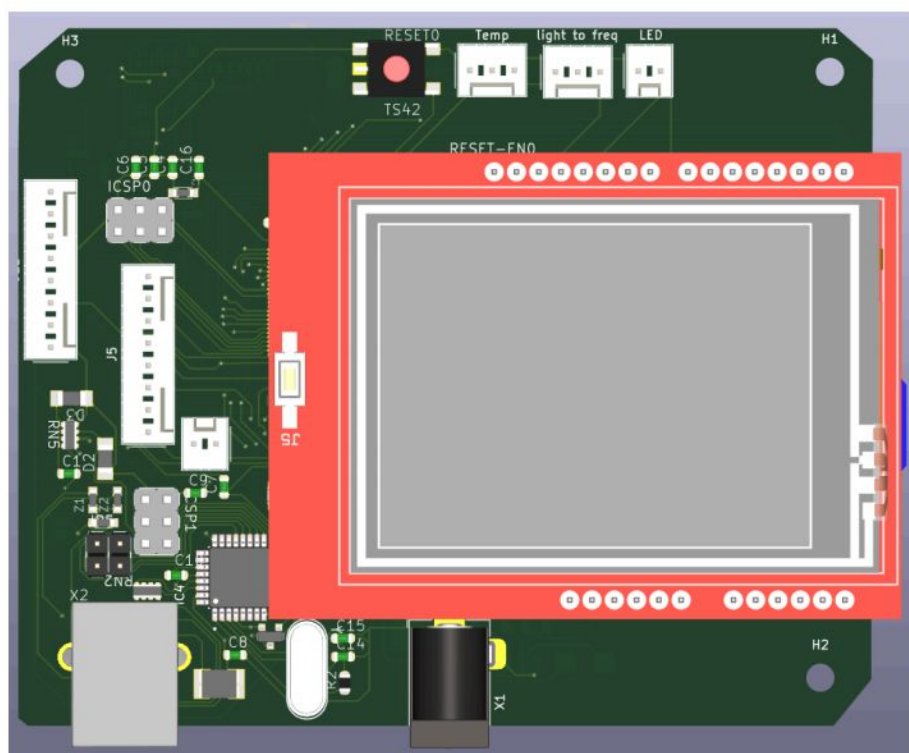


Fig. 18(B): 3D Model of PCB(B)

4.7 Testing and Calibration:

For accurate turbidity readings, the sensor is to be tested and calibrated. This involves submerging the sensor in standard solutions of known turbidity (in increasing order) as shown in Fig. 19, recording the corresponding frequencies, and plotting a graph of Turbidity vs Frequency. A curve is fit to the data to derive a calibration equation, which can then be used to determine the turbidity of unknown solutions.

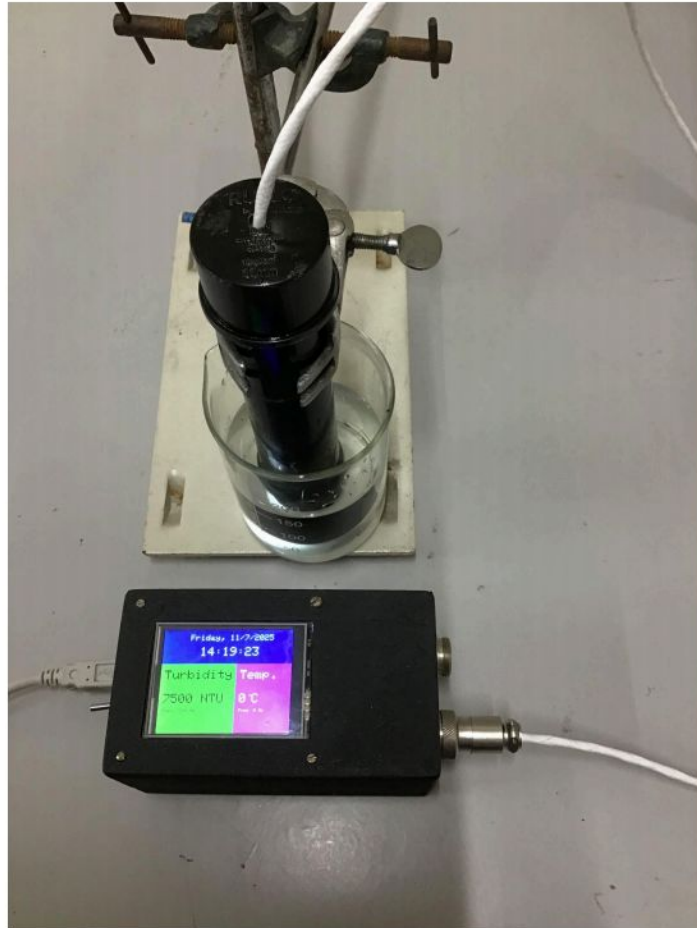


Fig. 19: Calibration

For the calibration of the sensors 7 solutions were used: DM water (0 NTU), 20(NTU), 40(NTU), 100(NTU), 200(NTU), 300(NTU), 400(NTU). On dipping the Turbidity Sensor in the solutions, in ascending order of turbidity, followed by descending order and observing the readings, it was observed that some external light/shadow is affecting the readings. The next test was done covering the beaker in black tape to simulate a darker environment and less effect of external light. Fig. 20 shows the observed readings and Fig. 21 shows the obtained graph.

turbidity(NTU)	freq(forward)(Hz)	Freq(backward)(Hz)	Freq(forw black)(Hz)
0	4520	4666	3145
20	7156	7763	2826
40	9520	12166	3420
100	8022	6709	3081
200	7272	6330	3686
300	6730	6813	4364
400	6269	6048	4398

Fig. 20: Table of Readings

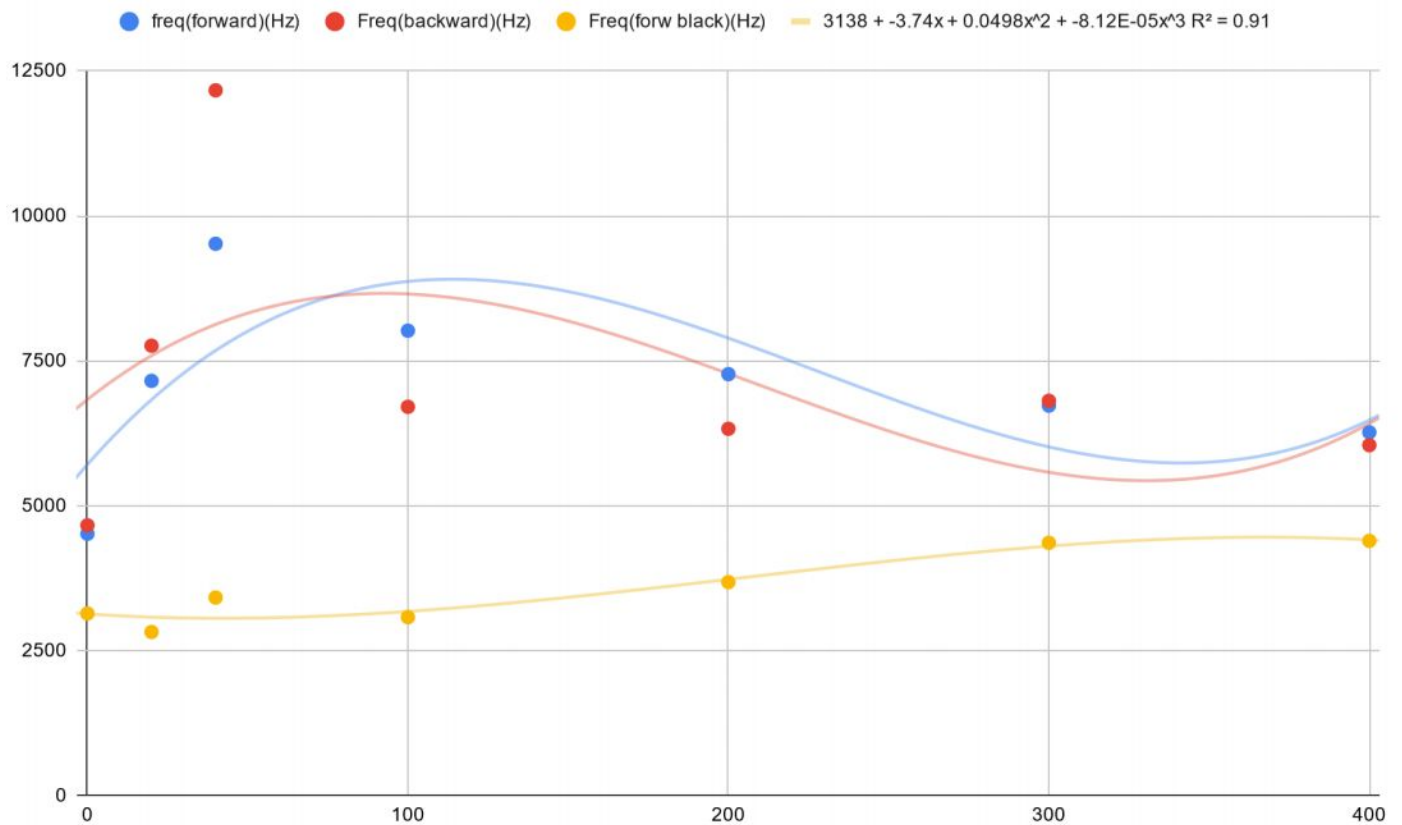


Fig. 21: Calibration Graph

ANALYSIS

5.1 Frequency Measurement using Arduino Mega and RTC:

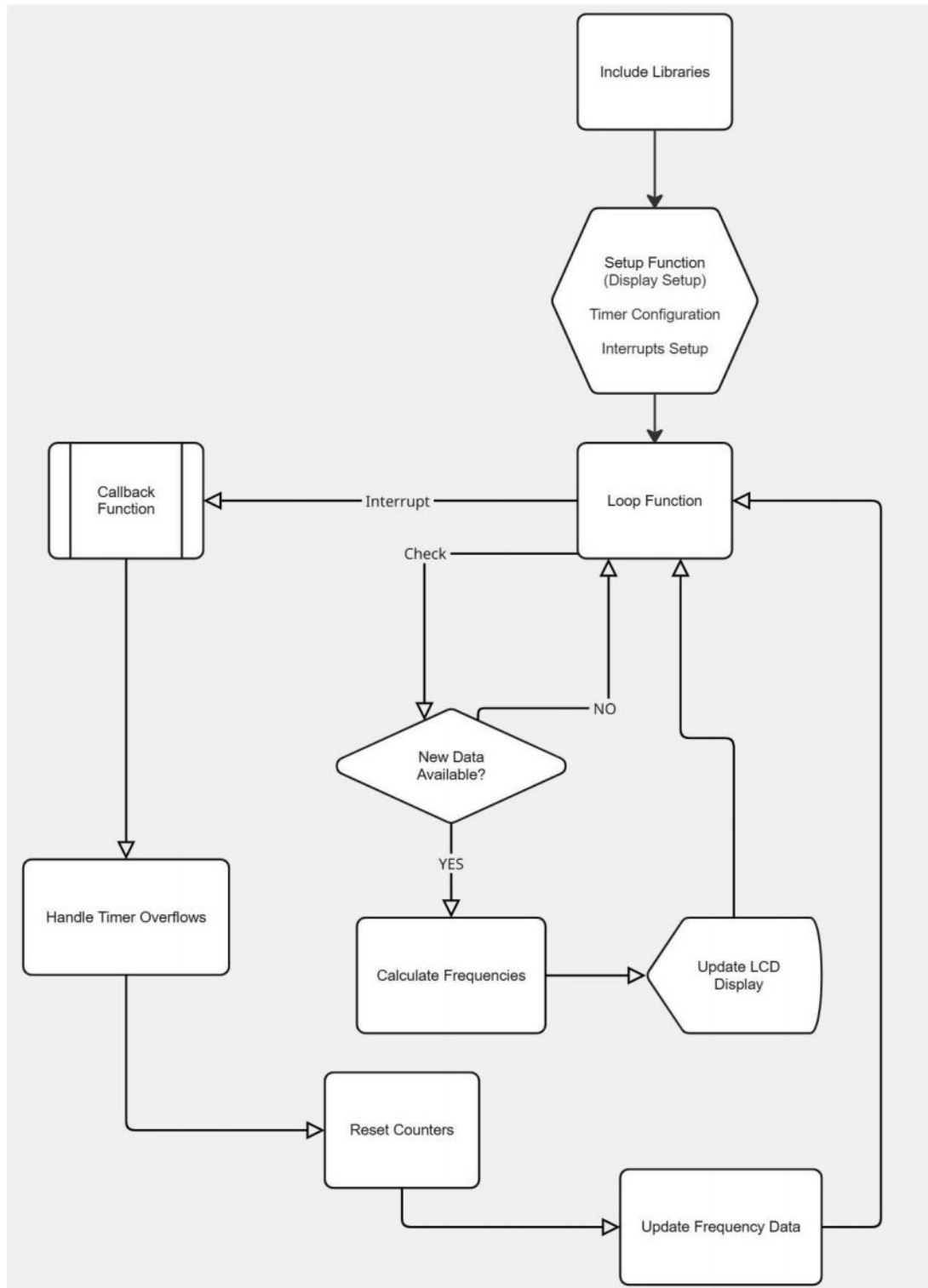


Fig. 22: Flowchart of the Frequency Logic

Timer Setup and Use:

The Flowchart in Fig. 22 explains the working of the code.

The Arduino Mega contains several 16-bit hardware timers (Timers 1, 3, 4, and 5), each set up to operate with a clock prescaler of 1024 (using CS12, CS11, CS10, etc.). They are thus configured to count external events or internal clocks slowly enough that they will not overflow too rapidly.

All these timers are independent and utilized to count edges of rising or falling square wave signals applied to various pins. Since timers are 16-bit, they can count up to 65535 before overflow. In order to compensate for overflows within a 1-second time period, interrupt service routines (ISR(TIMERx_OVF_vect)) are utilized to increment overflow counters.

RTC Synchronization:

The RTC's 1 Hz square wave output is attached to digital pin 18 (an external interrupt pin). This triggers the callback() function every second. This function:

- Prevents overflow interrupts to pause timer counts briefly.
- Takes a snapshot of the current timer values and their corresponding overflow counts.
- Clears the timers and overflow counters to zero.
- Re-enables the interrupts.
- Sets a flag newData = true to indicate the loop() that new frequency data is available.

Frequency Calculation

In the loop(), if newData is true, the following code is executed:

Multiplies the number of overflows by 65536 (maximum timer value) and adds the current timer count to obtain the cumulative pulse count per second.

This cumulative value is the frequency (pulses/second) for the given channel, as the RTC interrupt interval is 1 second.

The frequencies are sent to the serial monitor and displayed on a TFT LCD using the MCUFRIEND_kbv and Adafruit_GFX libraries.

5.2 Working Physics behind the Turbidity Sensor :

The **turbidity sensor** measures how cloudy or murky a liquid is. This cloudiness is caused by **suspended particles** (like silt, organic matter, or microorganisms) that **scatter light** passing through the liquid.

The Principle of Light Scattering:

When the LED shines through the liquid, the light travels and encounters suspended particles. These particles may be of various sizes and scatter light in all directions, which is a phenomenon known as **Mie Scattering**, when the particles are of similar size to the wavelength of the scattered light. The cleaner the water the lesser particles and hence lesser scattering, similarly more particles would mean more scattered light.

At **90°** to the incoming beam, the TSL235R detects the light scattered sideways (perpendicular to the LED beam). This angle is chosen because it minimizes interference from direct or refracted light and enhances sensitivity to scattered light.

The physics behind this is based on **Mie theory**, which is a solution to Maxwell's equations for the scattering of electromagnetic waves by spherical particles. For smaller particles, **Rayleigh scattering** applies, where scattering is inversely proportional to the 4th power of wavelength:

$$I \propto \frac{1}{\lambda^4}$$

For turbidity sensors, most particles fall into the **Mie scattering** range, where scattering intensity depends on particle size, shape, concentration, and wavelength.

The angular distribution of scattered light varies, but the **90°** position is a reliable compromise between sensitivity and noise.

CONCLUSION

The development of a compact, low-power turbidity sensor using open-source tools and accessible hardware platforms demonstrates the potential of combining modern electronics, 3D printing, and embedded systems for environmental sensing applications. This project integrates the TSL235R light-to-frequency converter with the Arduino Mega microcontroller to create a system that captures real-time turbidity data.

Through the use of the DS3231 Real-Time Clock module, synchronized sampling was achieved, ensuring consistent timing for frequency measurements. The 3D-printed enclosure was crucial in reducing environmental light interference and waterproofing the internal electronics. Displaying real-time data on both a TFT LCD and a Python-based GUI helped in effective monitoring and visualization.

The Design of a Compact PCB was also very useful to engage multiple channels and reduce costs.

During Calibration procedure it was found that a few tweaks to the sensor can give us more accurate results, the improvements for which are ongoing.

Overall, this project has provided valuable hands-on experience in sensor integration, embedded systems design, frequency measurement, signal processing, and interface development. The work serves as a proof of concept for low-cost, customizable turbidity measurement systems that can be adapted for educational, environmental, or industrial purposes. Using open-source or free-to-use EDA Softwares and tools, such as Arduino IDE, KiCAD, Fusion 360, TinkerCAD, and Visual Studio Code, reducing the Costs significantly.

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